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DiffusionFF: A Diffusion-based Framework for Joint Face Forgery Detection and Fine-Grained Artifact Localization

Siran Peng[1,2*] Haoyuan Zhang[2,1*] Li Gao[3] Tianshuo Zhang[2,1]
Xiangyu Zhu[1,2] Bao Li[1,2] Weisong Zhao[4] Zhen Lei[1,2,5,6†]
[1]MAIS, Institute of Automation, Chinese Academy of Sciences
[2]School of Artificial Intelligence, University of Chinese Academy of Sciences
[3]China Mobile Financial Technology Co., Ltd. [4]IIE, Chinese Academy of Sciences
[5]CAIR, HKISI, Chinese Academy of Sciences [6]SCSE, FIE, M.U.S.T
{pengsiran2023, zhanghaoyuan2023, zhen.lei}@ia.ac.cn

Abstract

The rapid evolution of deepfake technologies demands robust and reliable face forgery detection algorithms. While determining whether an image has been manipulated remains essential, the ability to precisely localize forgery clues is also important for enhancing model explainability and building user trust. To address this dual challenge, we introduce DiffusionFF, a diffusion-based framework that simultaneously performs face forgery detection and fine-grained artifact localization. Our key idea is to establish a novel encoder-decoder architecture: a pretrained forgery detector serves as a powerful "artifact encoder", and a denoising diffusion model is repurposed as an "artifact decoder". Conditioned on multi-scale forgery-related features extracted by the encoder, the decoder progressively synthesizes a detailed artifact localization map. We then fuse this fine-grained localization map with high-level semantic features from the forgery detector, leading to substantial improvements in detection capability. Extensive experiments show that DiffusionFF achieves state-of-the-art (SOTA) performance across multiple benchmarks, underscoring its superior effectiveness and explainability.

1. Introduction

The rapid advancement of deepfake technologies [12, 18, 29, 37] has enabled the creation of forged facial content with highly convincing perceptual quality, posing serious threats to information integrity and public trust. This escalating concern underscores the urgent demand for robust and reliable face forgery detection algorithms. Traditionally, research in this area focuses on the binary classification task of determining whether an image has been manipulated. However, there is a growing recognition of the need to precisely localize forgery clues. By pinpointing the manipulation traces within an image, artifact localization offers clear, human-interpretable evidence that supports the model's decisions, thereby improving explainability and fostering greater user trust. To simultaneously achieve both objectives, recent studies are increasingly exploring unified frameworks that integrate detection and artifact localization.

However, the precise localization of fine-grained forgery clues remains a significant challenge. Conventional mask-based methods [13, 17, 60] are inherently coarse, highlighting only broad regions of manipulation. To overcome this

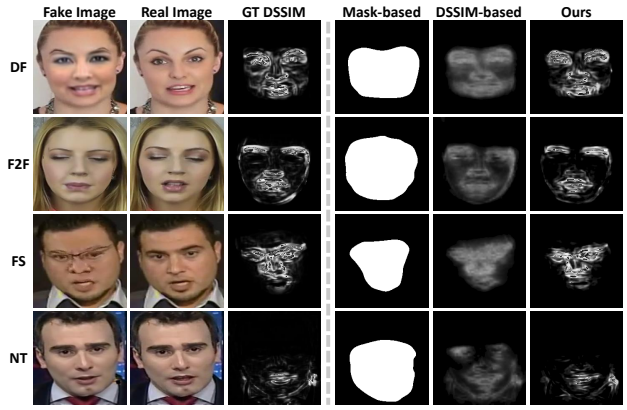


Figure 1. Visual comparison of DiffusionFF with mask-based and existing DSSIM-based artifact localization methods across four manipulation types (DF, F2F, FS, and NT) from the FaceForensics++ (FF++) dataset [35]. The Ground-Truth (GT) DSSIM map is generated by comparing each fake image with its corresponding real image using the algorithm detailed in Section 3.1.1.

*Equal contribution.
†Corresponding author.

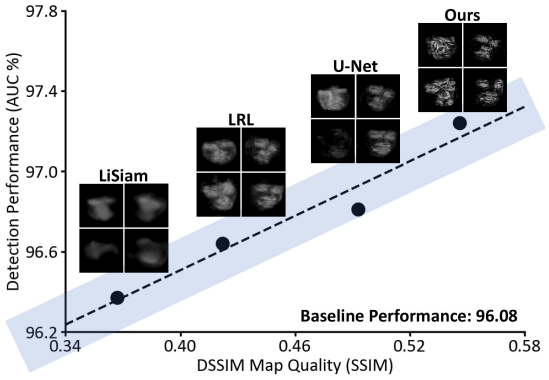


Figure 2. Correlation between the quality of the estimated DSSIM maps and detection performance. We fuse the DSSIM maps estimated by LiSiam [47], LRL [2], U-Net [34], and our DiffusionFF with high-level semantic features from a shared forgery detector to obtain classification results. Integrating estimated DSSIM maps into the detection network consistently improves performance, with higher-quality maps leading to greater performance gains.

limitation, several studies have explored using Structural Dissimilarity (DSSIM) maps [49] to guide localization map generation. By directly comparing pixel-level differences between aligned real and fake image pairs, DSSIM maps effectively capture subtle forgery artifacts. Despite this potential, existing DSSIM-based approaches [2, 47] are constrained by their reliance on direct regression frameworks, which tend to smooth over subtle manipulation traces, resulting in blurry and less informative localization maps. A visual comparison of these methods is presented in Figure 1.

In [47], the estimated DSSIM maps are reintegrated into the network to enhance detection capability. This strategy serves as a key motivation for our work. As shown in Figure 2, our analysis reveals a strong correlation between the quality of these maps and overall detection performance. Specifically, incorporating estimated DSSIM maps into the detection network consistently improves performance, with higher-quality maps leading to greater performance gains.

Considering these factors, we propose DiffusionFF, a novel framework designed to simultaneously tackle the dual challenges of face forgery detection and fine-grained artifact localization. The core innovation of DiffusionFF lies in its rational and intuitive encoder-decoder architecture, where a pretrained forgery detector functions as a powerful "artifact encoder", and a denoising diffusion model [11] is repurposed as an "artifact decoder". Conditioned on multi-scale forgery-relevant features extracted by the encoder, the decoder progressively estimates a detailed DSSIM map, which precisely localizes fine-grained forgery clues. This estimated DSSIM map is subsequently fused with high-level semantic features from the forgery detector, leading to substantial improvements in detection performance. In conclusion, the contributions of this paper are as follows:

- We establish a novel encoder-decoder architecture that leverages a pre-trained forgery detector as an "artifact encoder" and repurposes a denoising diffusion model as an "artifact decoder". Conditioned on multi-scale forgery-related features from the encoder, the decoder progressively synthesizes a fine-grained artifact localization map.
- We integrate the generated localization map with high-level semantic features from the forgery detector, resulting in significant improvements in detection capability.
- DiffusionFF achieves state-of-the-art (SOTA) detection performance and surpasses existing artifact localization methods in revealing manipulation traces, demonstrating its superior effectiveness, reliability, and explainability.

2. Related Works & Motivation

2.1. Face Forgery Detection

2.1.1. Detection-only Methods

Detection-only methods for face forgery detection primarily focus on achieving high accuracy, typically only providing a binary classification outcome. Existing studies can be grouped into four main categories [30]: spatial-domain, time-domain, frequency-domain, and data-driven approaches. Spatial-domain methods detect forgeries by analyzing image-level artifacts such as color mismatches [9], unnatural saturation [26], and visual anomalies [3, 54, 59]. Time-domain techniques model temporal dynamics across video frames to capture inter-frame inconsistencies [53, 57]. Frequency-domain methods apply algorithms like the FFT [40], DCT [32], and DWT [15, 21, 31] to expose subtle manipulation traces that are often imperceptible in the spatial or time domains. Lastly, data-driven approaches boost detection performance by leveraging advanced network architectures and optimized training strategies [14, 50, 51, 61].

2.1.2. Joint Detection and Localization Methods

In addition to detection-only approaches, some studies have developed unified frameworks capable of simultaneously performing face forgery detection and artifact localization. These methods can be broadly classified into two types: mask-based and DSSIM-based approaches. Mask-based methods treat artifact localization as a segmentation task, generating binary masks that indicate forged regions. These methods primarily exploit semantic features, analyze noise patterns, or leverage meta-learning techniques to improve segmentation precision [13, 17, 60]. In contrast, DSSIM-based approaches aim to produce fine-grained, pixel-level localization maps that reveal subtle traces of manipulation. To enhance localization quality, these approaches have explored attention mechanisms [2] or Siamese network architectures [47]. Notably, the development of these methods is facilitated by prominent datasets like FaceForen-

sics++ (FF++) [35] and DeeperForensics-1.0 [16], which provide the fully aligned real and fake image pairs necessary for generating Ground-Truth (GT) DSSIM maps. Unlike coarse binary masks, DSSIM maps offer a more precise and detailed representation of forgery artifacts. However, existing DSSIM-based methods rely on direct regression frameworks, leading to blurry and overly smoothed outputs.

2.2. Diffusion Models in Forgery Detection

In recent years, diffusion models [11] have emerged as a powerful class of generative tools, which iteratively transform noise into high-fidelity images by learning the underlying data distribution through a denoising process. In image forgery detection, notable works such as DiffForensics [56], InpDiffusion [48], and CBDiff [20] leverage diffusion models to create binary masks highlighting forged regions. In the specific domain of face forgery detection, DiffusionFake [39] employs a Latent Diffusion Model (LDM) [33] to reconstruct both the source and target identities from a single deepfake image. However, despite these advancements, the potential of diffusion models to synthesize fine-grained artifact localization maps remains largely unexplored.

2.3. Motivation

Recent studies are increasingly exploring unified frameworks that jointly address face forgery detection and artifact localization. Conventional mask-based methods provide only coarse localization of forged regions. On the other hand, while DSSIM maps can capture fine-grained manipulation traces, existing DSSIM-based approaches often produce blurry outputs due to their reliance on direct regression frameworks. To overcome these limitations and fully exploit the potential of DSSIM maps, we establish a novel encoder-decoder architecture that leverages a pre-trained forgery detector as an "artifact encoder" and reproposes a diffusion model as an "artifact decoder". Conditioned on forgery-relevant features from the encoder, the decoder progressively estimates a detailed DSSIM map, enabling precise localization of fine-grained artifacts. Motivated by our analysis in Figure 2, we fuse the estimated DSSIM map with high-level semantic features from the detector, leading to substantial improvements in detection performance.

3. Methodology

3.1. Preliminaries

3.1.1. GT DSSIM Map Formulation

To generate Ground-Truth (GT) DSSIM maps for supervising the training of denoising diffusion models and for evaluation purposes, we first extract frames and crop faces from both the original facial video and its manipulated counterparts, ensuring consistent settings across different video types. This process yields a set of fully aligned image pairs. For each image pair, we compute the DSSIM map by sliding a local square window across the two images and calculating the DSSIM value at each pixel location (i, j) as follows:

$$\text{DSSIM}(x, y) = 1 - \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}. \tag{1}$$

Here, $\text{DSSIM}(i, j)$ is the DSSIM value at the pixel position (i, j). The variables x and y represent the local windows centered at (i, j) in the original and manipulated images. μ_x and μ_y are the mean intensities of windows x and y, while σ_x and σ_y denote the variances of the corresponding windows. σ_{xy} represents the covariance between the two windows. The constants C_1 and C_2 are small values added to ensure numerical stability. Notably, real images are assigned pure black maps to represent zero dissimilarity.

3.1.2. Diffusion Model Training

We train our generative model following the Denoising Diffusion Probabilistic Model (DDPM) framework [11], which involves two primary processes: a forward diffusion process and a reverse denoising process. The forward process is a fixed Markov chain that progressively adds Gaussian noise to a clean image x_0 over T timesteps. This process has a key property that a noisy image x_t at any timestep t can be directly sampled from the original image x_0 as follows:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \tag{2}$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is a random noise sample. The noise level is dictated by a predefined variance schedule $\{\beta_t\}_{t=1}^T$. From this, we define $\alpha_t = 1 - \beta_t$ and the cumulative product $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$. The reverse process aims to train a denoising network $\epsilon_\theta(x_t, t)$ that predicts the noise ϵ added to the noisy image x_t. The training objective is to minimize the Mean Squared Error (MSE) between the predicted noise and the true noise, which can be expressed as follows:

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right]. \tag{3}$$

3.1.3. Diffusion Model Inference

During inference, the trained generative model reverses the forward diffusion process to reconstruct clean images from noisy inputs. Starting from a fully noisy image x_T, the model iteratively applies the reverse denoising process to refine it, eventually recovering the clean image x_0. At each timestep t, the denoising network $\epsilon_\theta(x_t, t)$ predicts the noise ϵ added to the image. The model then removes the predicted noise from the current noisy image x_t to produce a less noisy version x_{t-1}, using the following update rule:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t} \cdot z, \tag{4}$$

where $z \sim \mathcal{N}(0, \mathbf{I})$ is a random noise sample when $t > 1$. This process is repeated for each t from T down to 1, gradually refining the image until a clean sample is generated.

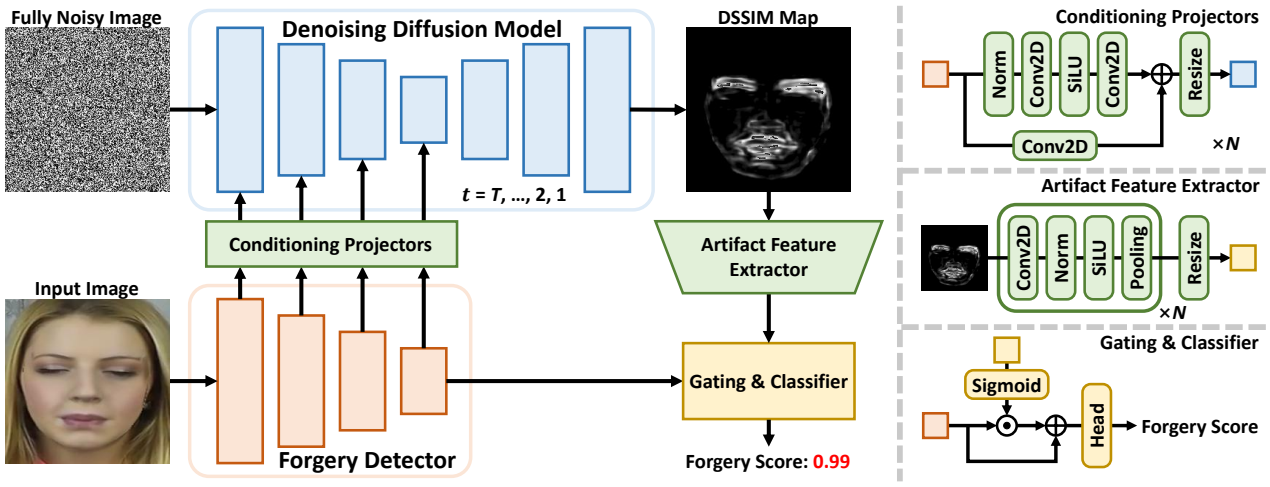


Figure 3. Overview of the DiffusionFF framework. Given an input facial image, DiffusionFF simultaneously predicts a forgery score and estimates a DSSIM map that precisely localizes fine-grained facial forgery clues. N denotes the number of stages in the forgery detector.

3.2. DiffusionFF

3.2.1. Overall Architecture

The proposed DiffusionFF framework simultaneously performs face forgery detection and fine-grained artifact localization, as illustrated in Figure 3. Given an input facial image, a pretrained forgery detector first extracts multi-scale features that encode forgery-related information. These features are then injected into the diffusion model via conditioning projectors to guide the generative process. Starting from random noise, the diffusion model progressively estimates a detailed DSSIM map that precisely localizes fine-grained forgery clues. An artifact feature extractor subsequently extracts informative features from the estimated DSSIM map. Finally, these artifact-aware features are integrated with high-level features from the detector through a gating mechanism to produce the classification outcome.

3.2.2. Conditional DSSIM Map Estimation via Diffusion

In previous studies, direct regression frameworks have been used to estimate DSSIM maps. However, such approaches often oversmooth subtle manipulation traces, resulting in blurry and less informative artifact localization. In contrast, diffusion models inherently support iterative refinement, allowing them to progressively enhance outputs and better capture fine-grained inconsistencies. Motivated by this advantage, we leverage a denoising diffusion model to estimate detailed DSSIM maps. To guide this generative process, we employ a pretrained forgery detector, which functions as a powerful "artifact encoder", to extract multi-scale features from the input facial image. Conditioned on these forgery-relevant features, the diffusion model is thus repurposed as an "artifact decoder", progressively synthesizing a fine-grained artifact localization map. Notably, we observe that jointly training the detector and diffusion model from scratch leads to training collapse, underscoring the importance of incorporating pre-learned forgery knowledge into the generation pipeline. To achieve this effective knowledge transfer, we design a set of conditioning projectors. These modules bridge the gap between the two models by aligning the spatial and channel dimensions of the detector's multi-scale features with the U-Net backbone [34] of the diffusion model. Once aligned, the forgery-related features are injected into the corresponding U-Net encoder stages, allowing the denoising diffusion model to fully exploit the detector's guidance and estimate detailed DSSIM maps.

3.2.3. Feature Fusion and Classification

Figure 2 demonstrates that integrating the estimated DSSIM maps into the detection network consistently improves performance. Therefore, we leverage the estimated high-fidelity DSSIM map to guide the final classification. Specifically, an artifact feature extractor first encodes the DSSIM map into artifact-aware features that capture subtle manipulation traces. These features are spatially and channel-wise aligned with the high-level semantic features from the final stage of the pretrained forgery detector. The two complementary feature sets are then fused via a gating mechanism, allowing the model to selectively emphasize manipulation-relevant regions. Finally, a classifier head produces the prediction. The overall process can be expressed as follows:

$$\text{Score} = \mathcal{H}\big(\sigma\left(\mathcal{E}(M_{DSSIM})\right) \odot F_{det} + F_{det}\big), \quad (5)$$

where M_{DSSIM} is the estimated DSSIM map, and F_{det} represents the detector's high-level features. $\mathcal{E}$ denotes the artifact feature extractor, σ is the Sigmoid activation, $\odot$ represents the Hardamard product, and $\mathcal{H}$ is the classifier head.

Table 1. Cross-dataset detection performance on the CDF2, DFDC, DFDCP, and FFIW benchmarks. Methods marked with $*$ are reproduced, while results for others are cited from the original papers. For each dataset, the best result is highlighted in **bold** and the second-best is underlined. Methods above the horizontal divider are detection-only, while those below are joint detection and localization approaches.

Method	Venue	Localization		Test Set AUC (%) ↑			
		Mask	DSSIM	CDF2	DFDC	DFDCP	FFIW
DCL [38]	AAAI 2022	✗	✗	82.30	-	76.71	71.14
SBI [36]	CVPR 2022	✗	✗	93.18	72.42	86.15	84.83
F^2Trans [27]	TIFS 2023	✗	✗	89.87	-	76.15	-
SeeABLE [19]	ICCV 2023	✗	✗	87.30	75.90	86.30	-
LAA-Net [28]	CVPR 2024	✗	✗	95.40	-	86.94	-
RAE [44]	ECCV 2024	✗	✗	95.50	80.20	89.50	-
FreqBlender [62]	NeurIPS 2024	✗	✗	94.59	74.59	87.56	86.14
UDD [7]	AAAI 2025	✗	✗	93.10	81.20	88.10	-
DFD-FCG [8]	CVPR 2025	✗	✗	95.00	81.81	-	-
Effort* [52]	ICML 2025	✗	✗	<u>95.73</u>	<u>84.78</u>	90.42	<u>88.53</u>
LRL* [2]	AAAI 2021	✗	✓	91.70	76.66	81.18	82.00
PCL+I2G [61]	ICCV 2021	✓	✗	90.03	67.52	74.37	-
LiSiam* [47]	TIFS 2022	✗	✓	90.36	72.59	82.06	76.52
AUNet [1]	CVPR 2023	✓	✗	92.77	73.82	86.16	81.45
Delocate [13]	IJCAI 2024	✓	✗	91.30	-	84.00	-
KFD [55]	ICML 2025	✓	✗	94.71	79.12	<u>91.81</u>	-
DiffusionFF (Ours)	-	✗	✓	**97.24**	**85.05**	**92.56**	**88.56**

3.2.4. Training Strategy

Given the significant differences between DSSIM map estimation and binary classification, jointly optimizing them often leads to suboptimal performance for both. To address this challenge, we propose a two-stage training strategy that fully decouples the two tasks at the training level. In the first stage, we freeze the pretrained forgery detector and focus on training the conditioning projectors and the denoising diffusion model. This stage is guided by the loss function defined in Equation 3, ensuring the model is effectively optimized for DSSIM map estimation. In the second stage, we freeze the forgery detector, conditioning projectors, and diffusion model, and instead optimize the artifact feature extractor alongside the classifier. This stage employs the standard Cross-Entropy (CE) loss to enhance classification performance. Our two-stage training strategy fully leverages the capabilities of each network component, ultimately maximizing the effectiveness of the DiffusionFF framework.

4. Experiments

4.1. Setup

4.1.1. Datasets

We perform model training and intra-dataset evaluation on the widely used FaceForensics++ (FF++) dataset [35], which contains 1,000 real facial videos and 4,000 fake ones generated by four manipulation techniques: DeepFakes (DF), Face2Face (F2F) [42], FaceSwap (FS), and NeuralTextures (NT) [43]. For cross-dataset evaluation, we select four popular benchmarks: Celeb-DeepFake-v2 (CDF2) [22], which applies advanced deepfake generation methods to YouTube celebrity videos; the DeepFake Detection Challenge (DFDC) [5] and its Preview version (DFDCP) [4], which contain videos with various perturbations such as compression, downsampling, and noise; and FFIW-10K (FFIW) [63], which incorporates multi-person scenarios.

4.1.2. Baselines

We compare our approach with sixteen SOTA baselines for face forgery detection, including ten detection-only methods: DCL [38], SBI [36], F^2Trans [27], SeeABLE [19], LAA-Net [28], RAE [44], FreqBlender [62], UDD [7], DFD-FCG [8], and Effort [52]; and six joint detection and localization methods: LRL [2], PCL+I2G [61], LiSiam [47], AUNet [1], Delocate [13], and KFD [55]. These baselines cover diverse technical paradigms, from traditional convolutions to SOTA Large Language Models (LLMs).

4.1.3. Evaluation Metrics

We evaluate detection performance using the Area Under the Receiver Operating Characteristic Curve (AUC), a

Table 2. Cross-dataset DSSIM map estimation performance. Note that the DFDC and DFDCP datasets lack GT DSSIM maps because their real and fake video pairs are not spatially aligned.

Dataset	Method	Evaluation Metrics			
		PSNR↑	SSIM↑	LPIPS↓	FID↓
CDF2	LiSiam [47]	21.985	0.367	0.464	256.201
	LRL [2]	20.801	0.422	0.455	258.674
	DiffusionFF	**30.697**	**0.546**	**0.376**	**98.982**
FFIW	LiSiam [47]	23.689	0.421	0.446	284.271
	LRL [2]	23.999	0.460	0.409	248.781
	DiffusionFF	**32.828**	**0.529**	**0.394**	**191.798**

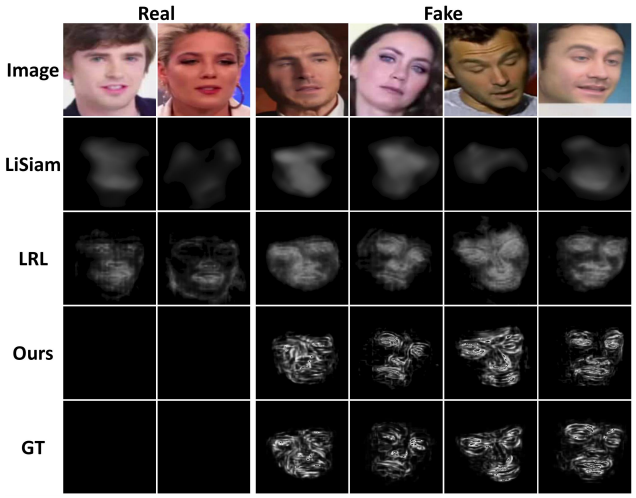


Figure 4. Qualitative DSSIM map estimation results on CDF2. DiffusionFF demonstrates strong generalization ability by producing more precise and fine-grained artifact localization results.

widely adopted metric for face forgery detection. Video-level results are reported by averaging frame-level predictions across each video. To assess the quality of the estimated DSSIM maps, we employ four standard metrics for evaluating image reconstruction and generation: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM) [49], Learned Perceptual Image Patch Similarity (LPIPS) [58], and Fréchet Inception Distance (FID) [10].

4.1.4. Implementation Details

For forgery detection, we adopt the preprocessing techniques, data augmentation strategies, and testing pipeline developed by SBI [36]. For artifact localization, we produce GT DSSIM maps on the FF++, CDF2, and FFIW datasets using the procedure outlined in Section 3.1.1. We employ a ConvNeXt-B network [24] pretrained on the FF++ dataset as the forgery detector, which consists of four hierarchical stages. For the denoising diffusion model, we use a U-Net

Table 3. Intra-dataset face forgery detection performance on the FF++ dataset. The proposed DiffusionFF framework yields the best overall results compared to existing SOTA approaches.

Method	Test Set AUC (%) ↑				
	DF	F2F	FS	NT	FF++
LRL [2]	**100**	**100**	99.95	99.59	99.88
PCL+I2G [61]	**100**	99.57	**100**	99.58	99.79
LiSiam [47]	**100**	99.99	99.99	99.46	99.86
AUNet [1]	**100**	99.86	99.98	99.71	99.89
RAE [44]	99.60	99.10	99.20	97.60	98.90
DiffusionFF	**100**	**100**	99.90	**99.86**	**99.94**

Table 4. Intra-dataset DSSIM map estimation performance on the FF++ dataset. Our method demonstrates superior performance compared to existing DSSIM-based localization approaches, with particularly significant improvement observed in the FID metric.

Method	Evaluation Metrics			
	PSNR↑	SSIM↑	LPIPS↓	FID↓
LiSiam [47]	20.753	0.408	0.426	292.149
LRL [2]	23.608	0.595	0.326	241.625
DiffusionFF	**26.376**	**0.718**	**0.198**	**43.093**

architecture [34] enhanced with a timestep encoder. Our two-stage training strategy is configured as follows. In the first stage, we train the conditioning projectors and the diffusion model for 100 epochs using the AdamW optimizer [25], with a batch size of 96 and an initial learning rate of 1×10^{-4}, which follows a cosine decay schedule. The total number of diffusion steps is set to $T = 50$. In the second stage, the artifact feature extractor and the classifier are trained for 5 epochs using AdamW, with a batch size of 128 and a fixed learning rate of 5×10^{-5}. All experiments are implemented in the PyTorch framework and executed on a cluster of eight NVIDIA RTX 3090 GPUs. For additional details, please kindly refer to the supplementary material.

4.2. Performance

4.2.1. Cross-Dataset Evaluation

Cross-dataset evaluation is a crucial protocol for assessing the generalization ability of face forgery detection models on previously unseen data. As shown in Table 1, we compare our method against recent SOTA approaches for face forgery detection. The proposed DiffusionFF framework outperforms all competing models, achieving the highest AUC scores across all unseen datasets, demonstrating its exceptional generalization capability. For artifact localization, we conduct both quantitative and qualitative comparisons with existing DSSIM-based techniques. The results

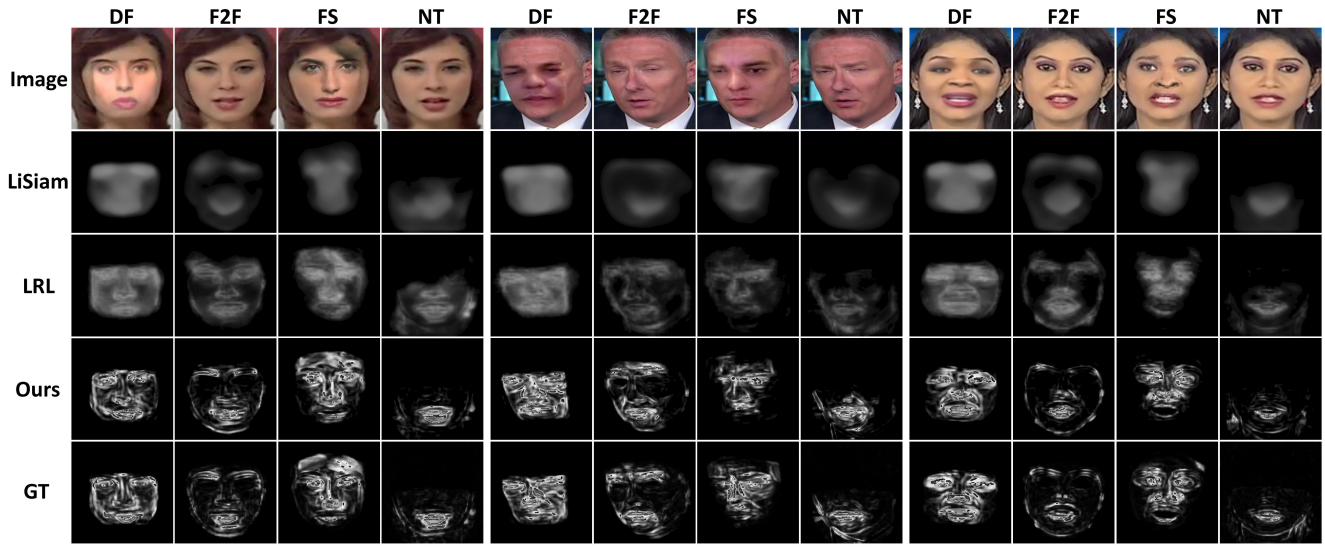


Figure 5. Qualitative DSSIM map estimation results on the FF++ dataset. Our method achieves the most visually faithful outcomes.

Table 5. Ablation study on the diffusion model. Our method produces the highest-quality DSSIM maps, validating the effectiveness of our design choices. "Cond." is short for conditioning.

Method	Evaluation Metrics			
	PSNR↑	SSIM↑	LPIPS↓	FID↓
Direct Regression	24.341	0.584	0.252	172.213
Latent-Space Diffusion	24.228	0.680	0.217	56.384
Final-Stage Cond.	26.154	0.711	0.204	43.812
Decoder Cond.	26.253	0.716	0.202	46.329
The Proposed	**26.376**	**0.718**	**0.198**	**43.093**

Table 6. Ablation study on the fusion module. The selected gating mechanism surpasses alternative fusion strategies in face forgery detection, confirming the soundness of our design choice.

Method	Test Set AUC (%) ↑				Avg.
	CDF2	DFDC	DFDCP	FFIW	
Addition	95.79	84.68	91.56	88.38	90.10
Hadamard Product	96.46	84.53	91.37	88.40	90.19
Concatenation	96.66	85.00	91.87	**89.01**	90.64
Cross-Attention	97.19	84.91	92.26	88.46	90.71
Gating Mechanism	**97.24**	**85.05**	**92.56**	88.56	**90.85**

in Table 2 and Figure 4 demonstrate the superior effectiveness of DiffusionFF in producing high-quality localization maps. This remarkable performance can be attributed to the novel application of diffusion models to estimate DSSIM maps, which enables the precise localization of fine-grained forgery clues and ultimately enhances detection capability.

4.2.2. Intra-Dataset Evaluation

In Table 3, we compare the intra-dataset detection performance of DiffusionFF with that of existing approaches, highlighting the superior discriminative power of our method. Furthermore, the quantitative results in Table 4 and qualitative outcomes in Figure 5 validate the efficacy of DiffusionFF in synthesizing high-quality localization maps.

4.3. Model Analysis

In this section, we present a comprehensive analysis of the DiffusionFF framework, including ablation studies on the diffusion model and fusion module, as well as evaluations of model generality and robustness. Additional ablation experiments, covering training and inference settings of the diffusion model, denoising seeds, and the two-stage training pipeline, are all detailed in the supplementary material.

4.3.1. Ablation Study on the Diffusion Model

We conduct a series of ablation experiments to evaluate the key design choices in our denoising diffusion model for fine-grained artifact localization. Specifically, our experiments investigate: (1) the effectiveness of iterative generation compared to direct regression; (2) the impact of utilizing diffusion models at the pixel level rather than in the latent space [33]; (3) the advantages of multi-stage conditioning over conditioning applied only at the final stage; and (4) the benefits of encoder-based conditioning relative to conditioning on the U-Net decoder. The quantitative DSSIM map estimation outcomes, as summarized in Table 5, substantiate the validity and soundness of our architectural choices.

Table 7. Evaluation of model generality. Integrating DiffusionFF as an auxiliary module consistently enhances the performance of existing forgery detectors, demonstrating its effectiveness as a plug-and-play improvement for face forgery detection models.

Method	Params	Test Set AUC (%) ↑			
		CDF2	DFDC	DFDCP	FFIW
EfficientNet-B4	19M	93.18	72.42	86.15	84.83
+ DiffusionFF	32M	95.54	82.19	89.03	86.37
Swin-B	88M	95.59	82.81	89.74	85.38
+ DiffusionFF	101M	96.93	84.54	90.78	87.01
ConvNeXt-B	89M	96.08	84.26	90.20	87.67
+ DiffusionFF	102M	**97.24**	**85.05**	**92.56**	**88.56**

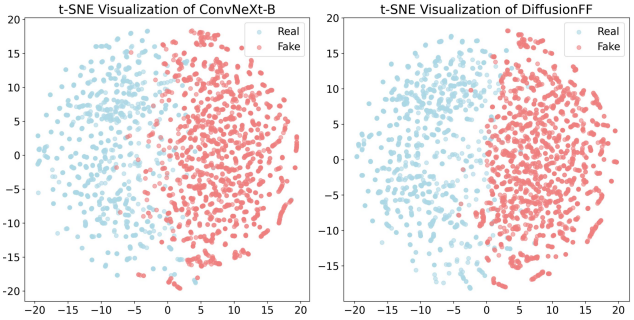


Figure 6. t-SNE visualizations of feature representations extracted by ConvNeXt and DiffusionFF on the CDF2 dataset. Compared to ConvNeXt, DiffusionFF yields more distinct and well-separated clusters, demonstrating its superior discriminative capability.

4.3.2. Ablation Study on the Fusion Module

To assess the effectiveness of the gating mechanism for feature fusion, we perform comparative experiments against several alternative strategies, including addition, Hadamard product, concatenation, and cross-attention [46]. As shown in Table 6, the cross-dataset detection results demonstrate the superior ability of the gating mechanism to seamlessly integrate artifact-aware and high-level semantic features.

4.3.3. Evaluation of Model Generality

We evaluate the generality of the DiffusionFF framework by examining its ability to enhance the detection performance of various detectors. Our study considers two CNN-based models, EfficientNet [41] and ConvNeXt [24], as well as the Transformer-based Swin model [23]. For each detector, we compare its baseline performance against its performance when augmented with DiffusionFF, which provides additional artifact localization maps serving as visual explanations. The Vision Transformer (ViT) [6] is excluded because its single-scale feature representation is incompatible with DiffusionFF's multi-scale conditioning

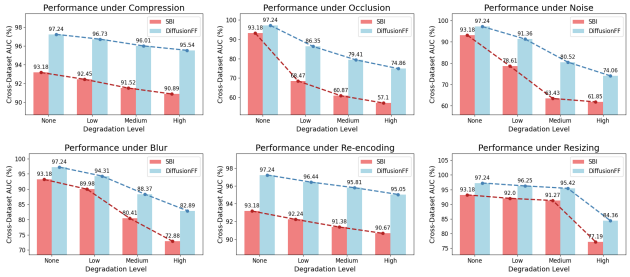


Figure 7. Evaluation of model robustness under various degradation conditions. DiffusionFF shows smaller performance declines than SBI [36] across all degradation types and severity levels.

mechanism. As shown in Table 7, all detectors exhibit consistent and substantial gains in cross-dataset detection performance when combined with DiffusionFF. Furthermore, t-SNE visualizations [45] in Figure 6 illustrate that DiffusionFF enables a clearer separation between real and fake samples than using the detector alone. These results confirm that integrating DiffusionFF as an auxiliary module significantly improves the performance of existing detectors, demonstrating its effectiveness as an explainable and plug-and-play enhancement for forgery detection models.

4.3.4. Evaluation of Model Robustness

We evaluate our model's robustness across six real-world degradations, each applied at three severity levels: JPEG compression (quality factors of 95, 85, and 75), random occlusion (masking 2%, 5%, and 10% of pixels), Gaussian noise (standard deviations of 0.01, 0.05, and 0.1), Gaussian blur (kernel/sigma values of 5/1, 7/2, and 9/3), repeated re-encoding (2, 3, and 4 iterations of JPEG compression with quality factors of 95, 90, and 85, respectively), and variable bitrate resizing (downsampling scales of 0.75, 0.50, and 0.25 with respective JPEG quality factors of 95, 90, and 85, followed by upscaling). As shown in Figure 7, the cross-dataset results on CDF2 demonstrate that DiffusionFF outperforms SBI [36] across all degradation conditions.

5. Conclusion

We propose DiffusionFF, a novel framework that simultaneously addresses the dual challenges of face forgery detection and fine-grained artifact localization. DiffusionFF introduces an innovative use of denoising diffusion models to estimate detailed DSSIM maps, enabling precise localization of fine-grained forgery clues and substantially improving detection capability. Extensive experiments demonstrate that our method not only achieves SOTA detection performance but also surpasses existing artifact localization approaches in revealing manipulation traces, underscoring its superior effectiveness, reliability, and explainability.

Acknowledgment

This work was supported in part by the Chinese National Natural Science Foundation Projects U23B2054, 62276254, the Science and Technology Development Fund of Macau Project 0140/2024/AGJ, the Science and Technology Foundation of Guangdong Province Project 2025A0505010025.

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